



KSI
KUMARAGURU
SCHOOL OF
INNOVATION

24ADC001- BUSINESS INTELLIGENCE

NEHA. N .K (24BAD078)



2026-2027
LAB RECORD

DEPARTMENT OF ARTIFICIAL
INTELLIGENCE AND DATA SCIENCE

INDEX

Expt. No	Title of the Experiment	Page No	Total Marks	Faculty Signature
1.	Explore the interface and basic features of the BI tool	3		
2.	Load and visualize a sample dataset.			
3.	Import a dataset into the BI tool. And Cleanse data by handling missing values, outliers, and inconsistencies.			
4.	Transform data to suit BI reporting requirements and Design a dashboard with key performance indicators (KPIs).			
5.	Develop interactive dashboards for dynamic data exploration.			
6.	Integrate data from various sources for comprehensive analysis			
7.	Implement advanced chart types (treemaps, heatmaps, etc.).			
8.	Apply BI tools for forecasting and predictive analytics.			

Average marks

Faculty Signature

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is the purpose of Business Intelligence (BI) tools like Tableau and Power BI?

Business Intelligence tools such as Tableau and Power BI are used to collect, analyze, visualize, and present business data through interactive dashboards and reports, enabling organizations to make informed decisions.

2. List any three common data sources that Tableau and Power BI can connect to.

Tableau and Power BI can connect to many types of data sources. Three common ones are:

- **Microsoft Excel** spreadsheets (.xlsx)
- **CSV (Comma-Separated Values)** files
- **SQL databases** such as MySQL, Microsoft SQL Server, or PostgreSQL

They can also connect to cloud services like Google Analytics, Azure, and Salesforce.

3. What is a dashboard in BI, and how is it different from a single chart?

A dashboard is a collection of multiple visualizations, charts, tables, and KPIs displayed on a single screen to provide an overall view of business performance. A single chart shows only one aspect of the data, while a dashboard combines different visuals and allows users to interact with the data for deeper analysis.

4. Define the following terms:

- a) **Visualization** - Visualization is the process of presenting data in graphical forms such as charts, graphs, maps, or dashboards to make information easier to understand and analyze.
- b) **KPI** - A KPI is a measurable value that indicates how effectively an organization is achieving its business objectives. Examples include sales revenue, profit, customer satisfaction, and inventory turnover.
- c) **Data Filter** - A data filter is a feature that displays only data that meets specific conditions, helping users focus on relevant information while hiding unnecessary records.

5. Why is it important to clean and transform data before visualization?

Cleaning and transforming data improves its quality by removing duplicates, correcting errors, handling missing values, and converting data into the required format. This ensures that visualizations are accurate, reliable, and meaningful, leading to better business decisions.

6. What kind of chart would you use to analyze stock price trends over time?

A **Line Chart** is the most suitable chart for analyzing stock price trends over time because it clearly shows how prices increase or decrease across different dates. It helps identify trends, fluctuations, and patterns in stock performance.

7. Explain the role of slicers in Power BI and filters in Tableau.

Slicers in Power BI and **filters in Tableau** allow users to interactively display only the required data based on selected criteria, such as date, region, or product category. They make dashboards more dynamic and help users analyze specific portions of the data without changing the original dataset.

8. What does a calculated field or measure do in BI tools?

A **calculated field** or **measure** performs calculations on existing data to create new values or metrics. It is used to calculate totals, averages, percentages, profit margins, growth rates, and other business metrics without modifying the original data.

9. What are the advantages of using interactive dashboards for financial data analysis?

Interactive dashboards allow users to monitor financial performance in real time, compare trends, analyze profits and losses, and quickly identify important insights. They also provide filtering, drill-down, and dynamic reporting features, making financial analysis faster and more effective.

10. Which function would you use to find the absolute difference between predicted and actual stock prices?

- The **ABS()** function is used to find the absolute difference between predicted and actual stock prices. It returns the positive value of the difference regardless of whether the prediction is higher or lower than the actual value.
- **Formula:**
$$\text{Absolute Difference} = \text{ABS}(\text{Predicted Price} - \text{Actual Price})$$
- This helps measure prediction errors without considering their direction.

IN-LAB EXERCISE:

Objective:

To understand the basic interface, data import process, and fundamental visualization capabilities of **Tableau** and **Power BI** using a stock prediction dataset, and to explore their usefulness in analyzing financial data.

Tools Required:

- Tableau Public / Tableau Desktop
- Power BI Desktop
- Sample Dataset (CSV/Excel) containing:
Date, Stock Name, Open, Close, High, Low, Volume, Predicted Price

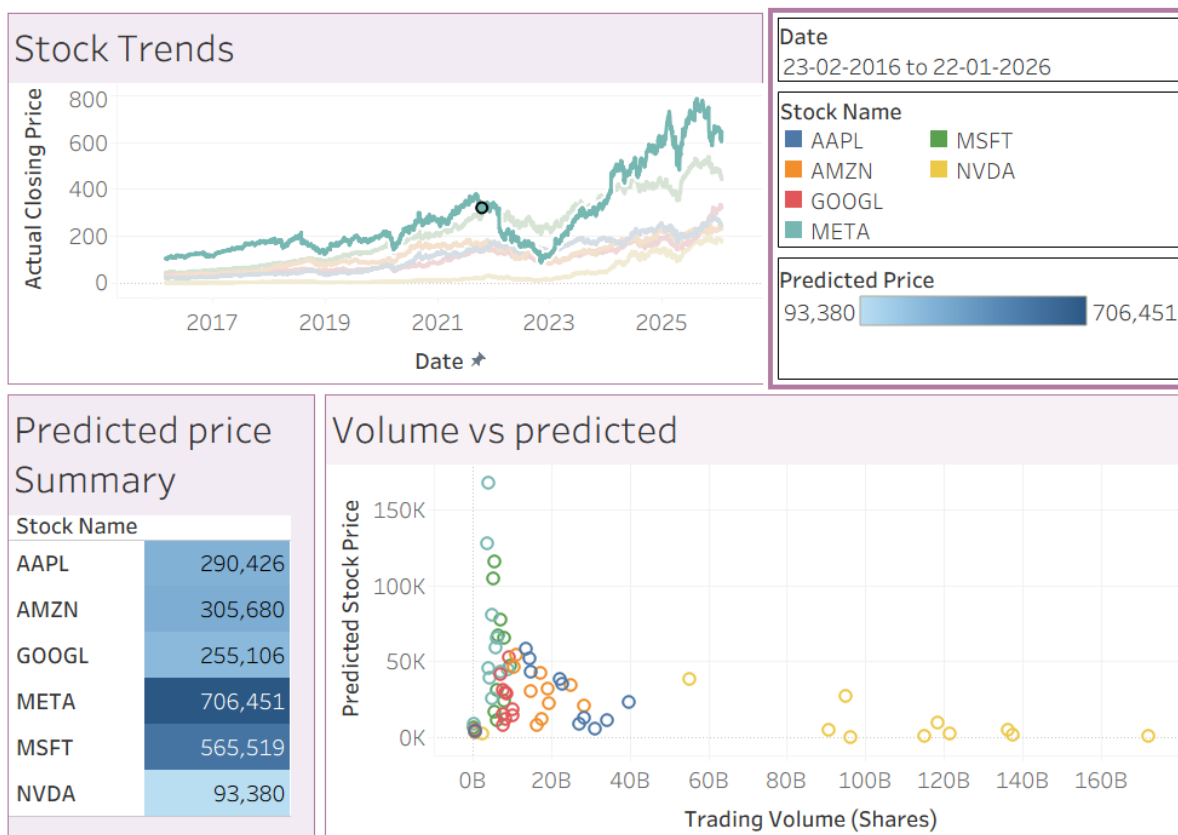
Exercise Steps:

Part 1: Working with Tableau

1. **Launch Tableau** and connect to the stock prediction dataset.
2. **Explore the Interface:**
 - Understand components like **Data Pane**, **Sheets**, **Marks Card**, and **Dashboards**.
3. **Create Basic Visualizations:**
 - **Line Chart:** Show Close Price over time.
 - **Scatter Plot:** Plot Volume vs. Predicted Price.
 - **Highlight Table:** Display Predicted Price by stock.
4. **Apply Filters:**
 - Add filters for Stock Name and Date Range.
5. **Create Calculated Field:**
 - Calculate prediction error as $|\text{Predicted} - \text{Close}|$.
6. **Design a Basic Dashboard:**
 - Combine your charts into one interactive dashboard.

OUTPUT:

FAANG Stock Performance & Prediction Analytics



1. Stock Trends Worksheet

- **Chart Used:** Line Chart.
- Maps the historical path of stock closing prices across a continuous timeline. Using distinct colours for each company makes it easy to compare their growth trajectories and long-term market trends immediately.

2. Predicted Price Summary Worksheet

- **Chart Used:** Highlight Table (Heatmap Matrix).
- Organizes the aggregate predicted stock values into a clean text grid. It applies a shading gradient where darker blue cells represent higher values, enabling the reader to instantly spot the highest-valued stock predictions without reading individual digits.

3. Volume vs Predicted Worksheet

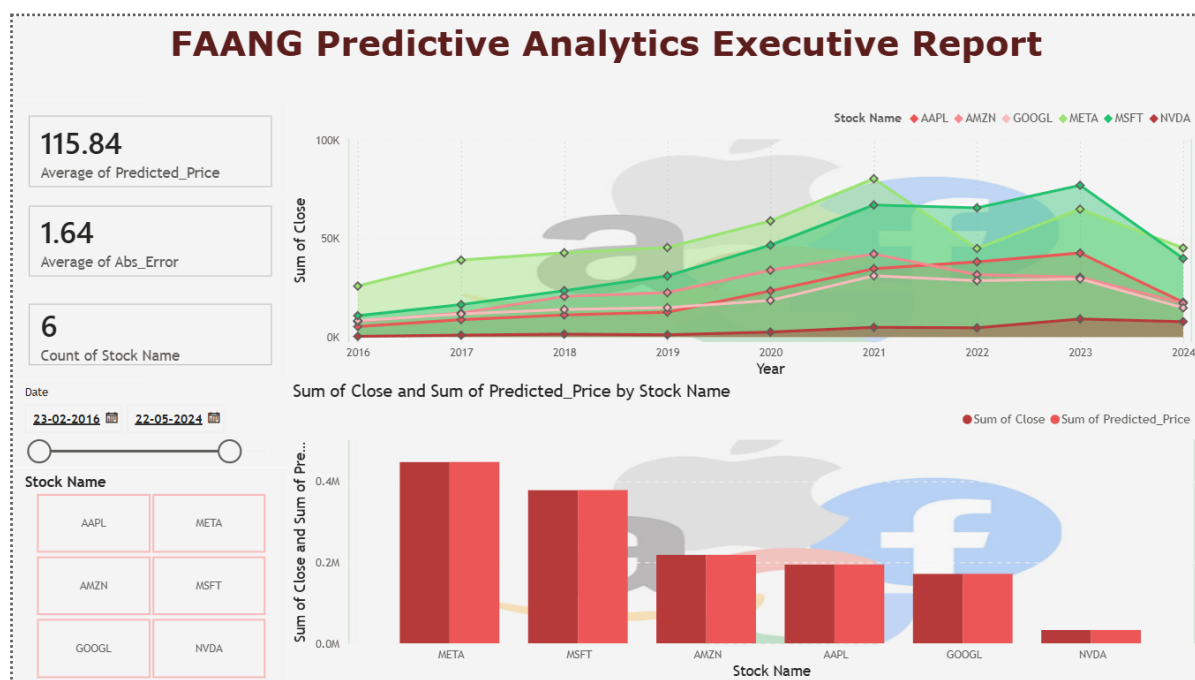
- **Chart Used:** Scatter Plot.
- Plots trading volume on the horizontal axis and predicted stock prices on the vertical axis. Distributing daily snapshots as scattered points shows the density and relationship between trading activity levels and forecasted values.

Part 2: Working with Power BI

1. **Open Power BI Desktop** and import the same dataset.
2. **Explore the Interface:**
 - Familiarize with **Fields Pane, Visualizations Pane, Report View, Data View.**
3. **Create Basic Visualizations:**
 - **Line Chart:** Trend of Close Price over time.
 - **Bar Chart:** Compare Predicted Price vs. Close Price.
 - **Card (KPI):** Display average prediction error.
4. **Use Filters and Slicers:**
 - Filter data based on Stock Name and Date.
5. **Create a Simple Measure** using DAX:
 - Example: $\text{Prediction Error} = \text{ABS}(\text{Predicted} - \text{Close})$
6. **Build a Report Page:**
 - Arrange visuals in a single interactive page.

OUTPUT:

<https://app.powerbi.com/groups/me/reports/093b53fb-918e-4bac-b252-ee158c222e9f/6fade837b3acb0d06e56?experience=power-bi>



1. KPI Cards (Top Left Panel)

Average Predicted Price (115.84):

Displays the average predicted stock price across the selected period, providing an overview of the model's expected market value.

Average Absolute Error (1.64):

Shows the average prediction error using $|\text{Predicted Price} - \text{Close Price}|$. A lower value indicates better prediction accuracy.

Count of Stock Name (6):

Indicates that the dashboard contains data for six different technology companies.

2. Multi-Line Chart – "Stock Closing Price Trend by Year" (Top Right)

Chart Type: Multi-Line Chart

Displays the closing price trends of all stocks from 2016 to 2024. Each colored line represents a different company.

Helps compare stock performance over time and identify growth or decline trends.

3. Clustered Column Chart – "Actual vs Predicted Closing Price by Stock" (Bottom Right)

Chart Type: Clustered Column Chart

Compares the actual closing prices with the predicted prices for each company using side-by-side columns.

Allows users to evaluate how closely the predicted prices match the actual market prices.

4. Interactive Controls & Slicers (Bottom Left)

Date Slicer:

Allows users to filter the dashboard by selecting a specific date range.

Stock Name Slicer:

Enables users to select one or more stocks (AAPL, AMZN, GOOGL, META, MSFT, NVDA). All charts and KPI cards update automatically based on the selected stock(s).

Provides interactive filtering for focused and dynamic financial analysis.

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. Describe your experience with the Tableau interface. What components did you interact with?

The Tableau interface was intuitive and easy to use. I interacted with components such as the **Data Pane**, **Rows and Columns shelves**, **Marks Card**, **Show Me panel**, **Filters**, and **Dashboard workspace**. These features helped me create and customize different visualizations efficiently.

2. How did you import the dataset into Tableau and Power BI?

- In **Tableau**, I clicked **Connect - Microsoft Excel (or Text File)**, selected the dataset, and loaded it into the workspace.
- In **Power BI**, I used **Home - Get Data - Excel/CSV**, selected the file, previewed the data, and clicked **Load** to import it for visualization.

3. What types of visualizations did you create in both tools? List at least two per tool.

Tableau:

- **Line Chart** to display stock closing price trends over time.
- **Scatter Plot** to analyze the relationship between trading volume and predicted stock price.
- **Highlight Table** to compare predicted prices across different stocks.

Power BI:

- **Multi-Line Chart** to compare stock closing price trends for different companies.
- **Clustered Column Chart** to compare actual closing prices with predicted prices.
- **KPI Cards** to display the average predicted price, average prediction error, and total number of stocks.

4. How did you calculate the prediction error, and how was it displayed in your dashboard?

- The prediction error was calculated using the formula:
- **Prediction Error = ABS(Predicted Price – Close Price)**
- A calculated field named **Abs_Error** was created. In the Power BI dashboard, the average prediction error was displayed using a **KPI card**, allowing users to quickly assess the overall accuracy of the prediction model.

5. Which filtering or slicing techniques did you apply, and what insights did they reveal?

A **Date Range slicer** and a **Stock Name slicer** were applied in both Tableau and Power BI. These interactive filters allowed the dashboard to display data for selected companies and specific time periods. They helped compare stock performance, identify price trends, and analyze prediction accuracy for individual stocks.

6. Compare the process of creating a line chart in Tableau vs. Power BI. What differences did you notice?
- In **Tableau**, creating a line chart involved dragging the **Date** field to the Columns shelf and the **Close Price** field to the Rows shelf. Tableau automatically generated the visualization.
 - In **Power BI**, the **Line Chart** visual was selected first, and then the required fields were assigned to the Axis, Values, and Legend sections. Tableau offered a more drag-and-drop approach, whereas Power BI required configuring the visual through its Visualizations pane.

7. How did you organize the dashboard in Tableau and Power BI? What key elements did you include?

The dashboards were organized to provide both summary information and detailed analysis. KPI cards displaying the average predicted price, prediction error, and stock count were placed at the top or side. The dashboards included line charts for trend analysis, clustered column charts or highlight tables for comparison, scatter plots for relationship analysis, and interactive filters for stock names and date ranges. This layout made the dashboards easy to navigate and interpret.

8. Which tool (Tableau or Power BI) did you find more user-friendly for financial data visualization, and why?

I found **Power BI** more user-friendly because it provides a familiar interface, easy integration with Excel datasets, and a wide range of built-in visuals and KPI cards. Tableau, however, offered greater flexibility and more advanced visualization options, especially for creating interactive charts and highlight tables.

9. What challenges did you face while creating visualizations or dashboards?

Some challenges included importing and formatting the dataset correctly, creating calculated fields for prediction error, selecting suitable chart types, and mainly **organizing the dashboard layout effectively was a difficult task**. Maintaining consistent formatting and ensuring that filters updated all visuals correctly also required careful attention.

10. How can the knowledge gained from this lab be applied to real-world financial data analysis?

The concepts learned in this lab can be applied to real-world financial analysis by monitoring stock performance, comparing predicted and actual prices, evaluating forecasting accuracy, and identifying market trends. Interactive dashboards help investors, financial analysts, and businesses make informed decisions by presenting complex financial data in a clear and easy-to-understand format.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		